Software Stream - Challenge Overview

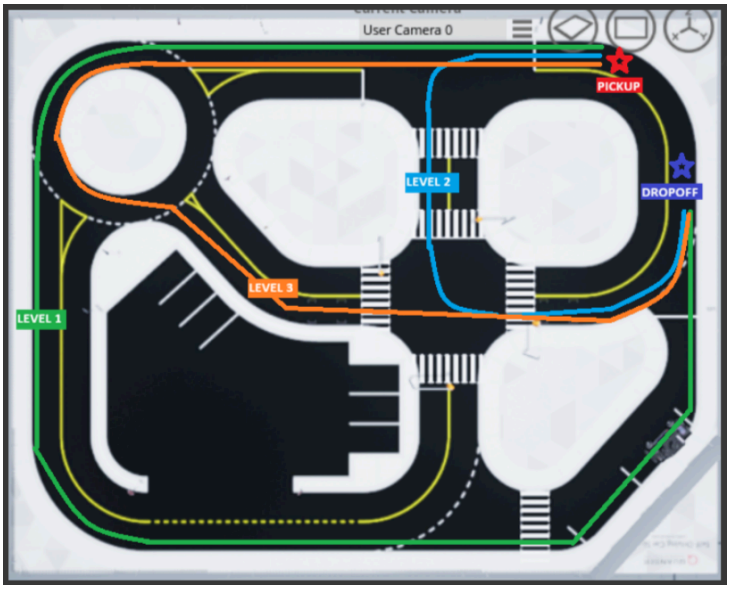
What is the Challenge?

The goal of this challenge is to introduce Matlab, Simulink and Qlab tools to students and for them to demonstrate problem solving, understanding, creativity and an engineering mindset **rather than** focusing on the quality of the code and algorithms.

Students are tasked with:

- Teams are an autonomous taxi services
- They must plan a route to pick up and drop off their passenger
- Team will pick 1 of the 3 levels and develop logic and algorithms using Qlabs, Matlab & Simulink for that route
- Explore core principles such as:
    - Path planning and behaviour
    - Perception
    - Localization
    - Speed & Controls
- Present their product and a technical overview to a panel of judges
    - 10 - 15 minute presentation
    - 5 minute Q&A session

Route Choices: You will not be scored differently based on what level you pick - But success in higher levels will be much more impressive!



Level 1- Challenge:

- Stop sign

Level 2 - Challenge:

- Stop sign
- Traffic light
- Create a custom path

Level 3 - Challenge:

- Stop sign
- Traffic light
- Pedestrian
- Create a custom path

<u>The winners of this Presentation challenge will be awarded $250. Other awards can be won.</u>

UWAFT Innovation Challenge
Hack the MOVE

Submission Naming conventions:

Ensure you submit everything in your team's google drive folder!

Presentation	STREAM2026_TEAMNAME_PRESENTATION Eg. Software2026_UWAFT_Presentation
Video	STREAM2026_TEAMNAME_VIDEOtype Eg. Software2026_UWAFT_Video Eg. Software2026_UWAFT_VideoDemo Eg. Software2026_UWAFT_VideoRulesoftheRoad
Technical files	STREAM2026_TEAMNAME_General content Eg. Software2026_UWAFT_Propulsion and control_code Eg. Software2026_UWAFT_Propulsion and control_block Eg. Software2026_UWAFT_Overallcode
Images (if needed)	STREAM2026_TEAMNAME_Image_Content Eg. Software2026_UWAFT_Propulsion and control_codeimage

Presentation Content

Remember: Judges prioritize engineering reasoning, system thinking, and justification over full completion

P1. Introduction (14 ots)

P1.1 - Elevator Pitch

- Start with a sharp, structured Elevator Pitch! - Make judges understand your system in 60-90 seconds
- Your team should Include:
    - Who you are (team name + quick intro)
    - What your autonomous taxi does
    - What makes your approach unique
    - What level/ path you chose (L1, L2, L3) and why

Helpful hint: Don't just say what you build - explain your design philosophy

P1.2 - Clearly define system failures

- Identify at least 2 points of failure and/ or failure modes :
    - Failure point - points of failures that could occur in your car (physical or software)
        - Eg. Wheels
        - Eg. Communication between two controllers

        - Eg. In software - if we don't process the data received from the sensors
        - If something is covered by snow - Detection algorithm is your failure point
    - Failure mode (general) - is how it fails
        - If there is snow on your stop sign
        - Delayed obstacle response
- Explain:
    - Why does this matter
    - How does it affect safety or performance?
    - Example: Inconsistent lane detection causes oscillatory steering behaviour, which impacts passenger comfort and increases risk and higher speeds.

Helpful hint: Judges want engineering insight, not just "we made it drive."

P1.3 - Present Clear, Logical Solutions

- Come up with your solutions
    - Connect each solution to an identified problems
    - Explain what you would/ have implement
    - Why did you choose that approach
    - What trade offs exists

Helpful hint: Judges are scoring logical reasoning, innovation, user benefit. Try to link it back to safety, reliability, passenger comfort, traffic law compliance.

P2. Core Principle execution & Justification (70 pts)

P2.1 - Demo Video (NO AUDIO)

- In this video the car must complete at least one full loop of your chosen path
- It is suggested to have text and pop ups to explain what is going on in your video
- Your team is also strongly suggested to talk while it drives. Be detailed!
    - Example: "At this point, our perception module identifies the lane boundaries. Our controller adjustes steering using proprotial gains to minimize cross track error."

Helpful hint: Judges are looking at understanding of core principles, clear explanation, confidence in system logic

P2.2 - System Architecture and Subsystem Design

- Show a clear decomposition into subsystems (perception, localization, planning, controls) - this can be down with a short video, screenshots of your screen, diagrams etc
- Show:
    - A block diagram
    - Subsystems are clearly separated:
        - Perception
        - Localization
        - Planning

        - Controls
    - Comments/ Notes
- Explain:
    - Data Flow
    - Why did you choose this structure?
    - Why this modular design.
    - Example: “We separated perception from planning to ensure modular debugging and future scalability”

P2.3 - Perception and Interpretation

- Visuals:
    - Show relevant code
    - Include your diagram/ image of your relevant block
    - Ensure you have clear labels/ titles
    - Neatly organized sub-blocks part of the same system (demonstrate the flow within your subsystem - could be organized based on function)
- Explain
    - What sensor you used (camera? Simulated LiDAR?)
        - Why that choice?
    - Lane detection logic
    - Traffic sign recognition logic
    - Obstacle handling
- Discuss:
    - Trade offs
    - Limitations

P2.4 - Localization and Mapping

- Visuals:
    - Show relevant code
    - Include your diagram/ image of your relevant block
    - Ensure you have clear labels/ titles
    - Neatly organized sub-blocks part of the same system (demonstrate the flow within your subsystem - could be organized based on function)
- Explain:
    - How does the vehicle know where it is?
    - How it interacts with mapped features
    - For levels 2 or 3
        - Explain intersection logic
        - Stoplight decision flow
        - Pedestrian logic
- Discuss:
    - Trade offs

    - Limitations

Helpful hints: Judges want structured thinking, not guesswork

P2.5 - Path Behaviour Planning

- Visuals:
    - Show relevant code
    - Include your diagram/ image of your relevant block
    - Ensure you have clear labels/ titles
    - Neatly organized sub-blocks part of the same system (demonstrate the flow within your subsystem - could be organized based on function)
- Explain:
    - How did you choose your route?
    - How do you handle:
        - Lane keeping
        - Stops
        - Turns
        - Intersections
- Discuss:
    - Safety constraints
    - Traffic Rules
    - Why your approach is structured
    - Trade offs
    - Limitation

P2.6 - Speed & Controls

- Visuals:
    - Show relevant code
    - Include your diagram/ image of your relevant block
    - Ensure you have clear labels/ titles
    - Neatly organized sub-blocks part of the same system (demonstrate the flow within your subsystem - could be organized based on function)
    - Demonstrate Graphs (suggestions: speed vs. time, acceleration vs. time, stopping response)
- Explain:
    - Why you chose your top speed
    - How you tuned gains
    - How you ensured smooth stopping

P3. Learnings (10 pts)

- Demonstrate that testing was done

- Include:
    - What failed?
    - What improved through testing?
    - What would you fix next?
    - What real-world constraints you didn't account for?

Helpful hint: Judges are evaluating engineering mindset and rigor

Overall: Communication (6pt)

Slide quality:

- Ensure:
    - Clean architecture diagrams
    - Large readable fonts
    - High contrast colors
    - Code snippets only when relevant
    - Graphs clearly labeled

Presenter quality

- Tips:
    - Speak confidently
    - Define technical terms briefly
    - Avoid reading slides
    - Stay within time

Other Awards - this will also be evaluated during your presentation!

1. **Engineering Insight Award**
    a. The best team to identify various points/ modes of failure and come up with appropriate and effective solutions

2. **From code to control award**
    a. Awarded to the team with the best technical and software skills, organization, communications

3. **Best Communication Award**
    a. Awarded to the team with the most polished slides and confident delivery

4. **Rules of the Road Award**
    a. Awarded to the team with the best video (can be the demo video or another video!)
    b. Demonstrates the best drive, including:
        i. Smooth, realistic driving behaviour
        ii. Communication of key concepts and ideas